

Use Case UH-UC1: Accommodation Upload (Accommodation Specialist)

Primary Actor: Accommodation Specialist

Supporting Actors: UniHaven System

Trigger: A property owner submits accommodation details to Accommodation Specialist for listing.

Description:

Accommodation Specialist inspects the accommodation and uploads its details to UniHaven, including property name, property type, availability period, number of beds, number of bedrooms, price, and address. The UniHaven System calculates the latitude and longitude using the DATA.GOV.HK Address Lookup Service and saves the property data to the database. New Accommodation can be searched by University Members.

Preconditions:

PRE-1: Property to be uploaded has been inspected and deemed suitable for listing.

PRE-2: Accommodation Specialist is authenticated and logged into the system.

PRE-3: Property owner has provided all necessary details.

Postconditions:

POST-1: Accommodation details are stored in UniHaven.

POST-2: Latitude and longitude of the property are recorded.

POST-3: Accommodation is available for search by University Members.

Main Success Scenario:

1. Accommodation Specialist selects the option to add new accommodation.
2. System displays a form for entering accommodation details.
3. Specialist enters property details (name, type, availability period, number of beds, number of bedrooms, price, and address).
4. System prompts the Specialist to confirm the submission.
5. Specialist confirms the submission.
6. System performs format and type checks of the details.
7. UniHaven System queries the DATA.GOV.HK Address Lookup Service using the address of property to obtain latitude and longitude of the property.
8. UniHaven System saves the accommodation details and location data to the database.
9. System prompts the Specialist to show accommodation is successfully uploaded.

Alternative Flows:

***a.** At any time, System fails:

To support recovery and accurate uploading, ensure all entered property details can be recovered from any step of the scenario.

1. Specialist restarts System, logs in, and requests recovery of prior state.
2. System reconstructs the prior state.
 - 2a. System detects anomalies preventing recovery:
 1. System signals error to Specialist, records the error, and enters a clean state.
 2. Specialist starts a new accommodation upload.

4a. Specialist realised the details input error before the confirmation of submission.

1. System displays a confirmation prompt with the accommodation details.
2. Specialist cancels the submission action.
3. System returns the Specialist to the details input process.

6a. Specialist input details on the wrong field detected by the System.

1. System sends instant notice to the Specialist for retyping the entry.
2. System return to the input process of property details
3. Specialist re-enters the input details.

6b. Specialist realise the wrong detail input after the confirmation of submission which cannot detected by System

1. Specialist immediately removes the uploaded accommodation after the uploading process manually.
2. Specialist starts a new accommodation upload for correct property details.

7a. The Address Lookup Service fails to return a valid result.

1. System displays an error message and prompts the specialist to manually enter the latitude and longitude.
2. Specialist enters the latitude and longitude manually.
3. System continues the step

8a. Database's storage is full and fails in saving accommodation and location data for property.

1. System stop sending data to database
2. System sends instant notice to Specialist for noticing the storage of full database
3. System stop the uploading process of UniHaven System.
4. Specialist fixing the storage problem.
 - 1a. Storage size of database is expend
 - 2a. Accommodation removal.

Use Case UH-UC2: Accommodation Removal (Accommodation Specialist)

Primary Actor: Accommodation Specialist

Supporting Actors: UniHaven System

Trigger: Accommodation Specialist receives a request to remove an accommodation listing from UniHaven.

Description:

The Accommodation Specialist reviews the request for removal and deletes the accommodation details from the UniHaven System. The system confirms the removal and updates the available database of accommodations.

Preconditions:

PRE-1: Specialist is authenticated and logged into the system.

PRE-2: Accommodation to be removed exists in the UniHaven database.

PRE-3: A valid request for removal has been received from the property owner, University Member, or Accommodation Specialist management.

Postconditions:

POST-1: Accommodation details are deleted from the UniHaven database.

POST-2: Removed accommodation is no longer available for search by University Members.

POST-3: UniHaven System logs the removal for auditing purposes.

Main Success Scenario:

1. Accommodation Specialist selects the option to remove an accommodation.
2. Specialist selects the accommodation and clicks the "Remove" button.
3. System prompts the Specialist to confirm the removal.
4. Specialist confirms the removal.
5. System deletes the accommodation details from the database.
6. System logs the removal action for auditing.
7. System confirms to the Specialist that the accommodation has been successfully removed.

Alternative Flows:

***a.** At any time, System fails:

To support recovery and accurate uploading, ensure all entered property details can be recovered from any step of the scenario.

1. Specialist restarts System, logs in, and requests recovery of prior state.
2. System reconstructs the prior state.
 - 2a. System detects anomalies preventing recovery:
 1. System signals error to Specialist, records the error, and enters a clean state.
 2. Specialist starts a new accommodation removal.

3a. Specialist selects the wrong accommodation for removal and realises before confirmation of removal

1. System displays a confirmation prompt with the accommodation details.
2. Specialist cancels the removal action.
3. System returns the Specialist to the list of accommodations.

4a. Specialist realises wrong accommodation removal after confirmation of removal

1. Specialist starts a new accommodation upload for uploading wrongly removed accommodation
2. Specialist starts a new accommodation removal for removing the correct accommodation

Use Case UH-UC3: Cancel Reservation (Accommodation Specialist)

Primary Actor:

Accommodation Specialist

Supporting Actors:

UniHaven System

Trigger:

A reservation fails at the contract stage, or the Accommodation Specialist specialist needs to cancel a reservation for other reasons.

Description:

The Accommodation Specialist cancels a reservation in UniHaven, and the system updates the status of the accommodation.

Preconditions:

1. PRE-1: The Accommodation Specialist is authenticated and logged into the system.
2. PRE-2: A reservation exists and is in the "Pending" status.

Postconditions:

1. POST-1: The reservation is marked as "cancelled" in UniHaven.
2. POST-2: The accommodation available for reservation again in UniHaven.

Basic Flow:

1. The Accommodation Specialist selects the option to view active reservations.
2. The system displays a list of active reservations.
3. The specialist selects a reservation to cancel.
4. The system displays the cancellation confirmation page.
5. The specialist confirms the cancellation.
6. The system marks the reservation as canceled and the accommodation as available.
7. The system confirms the cancellation to the specialist.

Alternative Flows:

***a.** At any time, System fails:

To maintain database integrity, ensure all changes can be rolled back from any step of the scenario.

1. The Specialist restarts the System, logs in, and requests recovery of prior state.
2. System reconstructs the prior state.
 - 2a. System detects anomalies preventing rolling back:
 1. System signals error to Specialist, records the error, and enters a clean state.
 2. Specialist starts the new cancellation.

Use Case UH-UC4: Receive Notifications of Reservations and Cancellations (Accommodation Specialist)

Primary Actor: Accommodation Specialist

Supporting Actors: UniHaven System

Trigger: An University Member reserves or cancels an accommodation.

Description:

The UniHaven system alerts the Accommodation Specialist when a reservation is made or canceled by a University Member. The specialist is responsible for managing the contract signing process (for reservations) or updating internal records (for cancellations).

Preconditions:

1. PRE-1: The Accommodation Specialist is authenticated and logged into the system.
2. PRE-2: An University Member has either reserved or canceled an accommodation.

Postconditions:

1. POST-1: The Accommodation Specialist is notified of the reservation or cancellation.

Basic Flow:

1. An University Member reserves or cancels an accommodation (see UC-3: Reserve Accommodation or UC-4: Cancel Reservation).
2. The UniHaven system sends a notification to the Accommodation Specialist about the reservation or cancellation.
3. For reservations:
 - 3.1: The system updates the reservation status to "Pending".
 - 3.2: The Accommodation Specialist receives the notification and begins managing the contract signing process.
4. For cancellations:
 - 4.1: The system updates the reservation status to "Cancelled" and marks the accommodation as available.
 - 4.2: The Accommodation Specialist receives the notification and updates any internal records.

Alternative Flows:

- 2a.** The notification fails to be sent.
 1. The system retries sending the notification.
 2. If the notification still fails, the system logs an error and alerts the system administrator.

Use Case UH-UC5: Search and View Accommodation (University Member)

Primary Actor: University Member (Student)

Supporting Actors: UniHaven System

Trigger: An University Member wants to search for off-campus accommodation.

Description:

The University Member searches for accommodation based on criteria such as type, availability period, number of beds, number of bedrooms, price, and distance from a selected campus. The system displays a list of matching accommodations sorted by distance.

Preconditions:

PRE-1: The University Member is authenticated and logged into the system.

PRE-2: Accommodation details have been added to UniHaven.

Postconditions:

POST-1: A list of accommodations matching the search criteria is displayed.

POST-2: The accommodations are sorted by distance from the selected campus (from nearest to furthest).

Main Success Scenario:

1. The University Member selects the option to search for accommodation.
2. The system displays a search form with filters (type, availability period, number of beds, number of bedrooms, price, selected campus and distance from campus).
3. The member enters their search criteria.
4. The system queries the UniHaven database for matching accommodations.
5. The system calculates the distance of each accommodation from the selected campus using latitude and longitude.
6. The system displays a list of accommodations sorted by distance (from nearest to furthest).
7. The member selects an accommodation to view its details.
8. The system displays information for the selected accommodation.

Alternative Flows:

***a.** At any time, System fails:

To maintain database integrity, ensure all changes can be rolled back from any step of the scenario.

3. The University Member restarts the System, logs in, and requests recovery of prior state.
4. System reconstructs the prior state.
 - 2a. System detects anomalies preventing rolling back:
 3. System signals error to Member, records the error, and enters a clean state.
 4. Member starts a new search process.

6a. No accommodations match the search criteria.

1. System displays a message indicating no results were found.
2. The member adjusts the search criteria and repeats the search.

Use Case UH-UC6: Reserve Accommodation (University Member)

Primary Actor:

University Member (Student)

Supporting Actors:

UniHaven System, Accommodation Specialist

Trigger:

An University Member selects an accommodation to reserve.

Description:

The University Member reserves an accommodation, and the system notifies the Accommodation Specialist to manage the contract signing process.

Preconditions:

PRE-1: The University Member searches and filters for the accommodations and enters the accommodation detail page.

PRE-2: The accommodation is available for reservation.

Postconditions:

POST-1: The accommodation is marked as “Pending” in UniHaven.

POST-2: The Accommodation Specialist is notified of the new reservation.

POST-3: The new reservation is created in the University Member and Accommodation Specialist reservation page.

Basic Flow:

1. The University Member selects an accommodation to reserve.
2. The system prompts the member to input the contact number and reservation period.
3. The system displays the reservation confirmation page.
4. The member confirms the reservation.
5. The system marks the accommodation as “Pending”.
6. The system notifies the Accommodation Specialist of the reservation.
7. The system uploads the accommodation to the University Member and Accommodation Specialist reservation page.

Alternative Flows:

***a.** At any time, System fails:

To maintain database integrity, ensure all changes can be rolled back from any step of the scenario.

5. The University Member restarts the System, logs in, and requests recovery of prior state.
6. System reconstructs the prior state.
 - 2a. System detects anomalies preventing rolling back:
 1. System signals error to the member, records the error, and enters a clean state.
 2. The member starts a new reservation.

3a. The member cancels the reservation before confirming.

1. The system returns to the accommodation detail page.

4a. The accommodation is no longer available.

1. The system displays an error message indicating the accommodation is no longer available.
2. The member selects another accommodation to reserve.

Use Case UH-UC7: Cancel Reservation (University Member)

Primary Actor: University Member

Supporting Actors: UniHaven System, Accommodation Specialist

Trigger: An University Member cancels a reservation before the exchange of contracts.

Description:

The University Member cancels a previously made reservation. The UniHaven System notifies the Accommodation Specialist about the cancellation.

Preconditions:

PRE-1: The University Member is authenticated and logged into the system.

PRE-2: The reservation has not reached the contract signing stage at that time.

Postconditions:

POST-1: The accommodation is marked as available in UniHaven.

POST-2: The Accommodation Specialist receives a notification of the cancellation.

POST-3: The accommodation reservation is marked as “cancelled” in the University Member and Accommodation Specialist reservation page.

Main Success Scenario:

1. The University Member selects the option to view their reservations.
2. The system displays a list of the member’s reservations.
3. The member selects the reservation to cancel.
4. The system displays the cancellation confirmation page.
5. The member confirms the cancellation.
6. The system marks the accommodation as available.
7. The system confirms the cancellation to the member.
8. The system notifies the Accommodation Specialist of the success of cancellation.

Alternative Flows:

***a.** At any time, System fails:

To maintain database integrity, ensure all changes can be rolled back from any step of the scenario.

1. The University Member restarts the System, logs in, and requests recovery of prior state.
2. System reconstructs the prior state.
 - 2a. System detects anomalies preventing rolling back:
 5. System signals error to the member, records the error, and enters a clean state.
 6. The member starts a new reservation.

5a: The University Member decides not to cancel the reservation.

1. The system returns to the reservation list.

Use Case UH-UC8: Rate Accommodation (University Member)

Primary Actor: University Member (Student)

Supporting Actors: UniHaven System

Trigger: An University Member rates an accommodation after the end of the contract.

Description:

The University Member rates the accommodation on a 6 point scale of 0-5, and the system records the rating.

Preconditions:

PRE-1: The University Member is authenticated and logged into the system.

PRE-2: The member's contract for the accommodation has ended and the reservation is marked as "completed".

PRE-3: The University Member did not submit ratings on this accommodation.

Postconditions:

POST-1: The accommodation's rating is updated in UniHaven.

Basic Flow:

1. The University Member selects the option to rate an accommodation.
2. The system displays a list of accommodations the member has rented.
3. The University Member selects the accommodation to rate.
4. The system displays a rating form (a rating score from 0-5 and a comment area).
5. The member submits their rating.
6. The system updates the accommodation's rating in the database.
7. The system confirms the rating submission.

Alternative Flows:

***a.** At any time, System fails:

To maintain database integrity, ensure all changes can be rolled back from any step of the scenario.

7. The University Member restarts the System, logs in, and requests recovery of prior state.
8. System reconstructs the prior state.
 - 2a. System detects anomalies preventing rolling back:
 7. System signals error to the member, records the error, and enters a clean state.
 8. The member starts a new rating.

5a. The University Member does not submit a rating.

1. The system returns to the list of accommodations.

5b. The University Member submits without selecting a rating score.

1. The system displays a message indicating that a rating score must be selected.

Note:

There are three status of reservation: "Pending", "Confirmed", "Cancelled" and "Completed". "Pending" refers to the stage when a contract is not signed, and "Confirmed" refers to the stage when the contract is signed. "Cancelled" means the reservation is cancelled, and "Completed" means the rent has finished and the reserved contract period has expired.